Classification of Music Based On Convolutional Neural Networks

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***Abstract*—Music genre classification is a fundamental and challenging task in music information retrieval. Traditional methods often rely on manual feature engineering combined with general classifiers, which have limitations in capturing complex hierarchical features of music. To address this issue, this study proposes a music genre classification approach based on Residual Network and validates it using the GTZAN dataset. First, log-mel spectrogram features are extracted from raw audio data using the LibROSA tool, and the features are preprocessed into a 3-channel image-like format to adapt to the input requirements of ResNet. Then, a classification model based on ResNet18 is constructed, with improvements including optimizing the fully connected layer structure, introducing dropout regularization, and adopting the AdamW optimizer with dynamic learning rate adjustment. The model is trained for 20 epochs with a batch size of 32. Experimental results show that the proposed model achieves an overall classification accuracy of 60.00% on the GTZAN dataset. Among them, the classical genre achieves perfect classification performance , while the country and metal genres also obtain relatively high F1-scores. However, the classification performance for stylistically similar genres such as jazz and rock is relatively limited due to overlapping temporal and spectral features. This study demonstrates that ResNet can effectively extract discriminative features from music audio spectrograms, providing a viable solution for music genre classification while reducing reliance on manual feature engineering. The research findings also offer valuable insights for further optimizing music classification models and improving classification accuracy for similar genres.**

***KeyWords*—music genre classification,Residual Network, log-mel spectrogram,deep learning**

I. INTRODUCTI

Convolutional neural networks have a history dating back to the 1980s, but their widespread adoption as a preferred method for various classification tasks has only occurred in recent years. The seminal work by Krizhevsky et al.[1] in 2012 marked a pivotal moment in large-scale visual recognition, and since then, CNNs have revolutionized pattern recognition by replacing manually engineered feature-based techniques. Their success has extended beyond visual domains, with effective applications in speech processing [3], environmental sound classification [2], and even early explorations in audio analysis [5]. These advancements highlight that models leveraging data locality, a key characteristic of CNNs, can offer robust solutions across diverse non-visual domains. A comprehensive overview of deep learning applications in audio-related tasks can be found in recent specialized literature [4].

Music genre classification stands as a critical and challenging task in music information retrieval. The complexity arises from the subtle distinctions between numerous genres, the existence of hybrid genres that blur categorical boundaries, and the inherent diversity of musical structures. Traditionally, music genre classification has relied on general classifiers—such as Gaussian mixture models, support vector machines , and hidden Markov models—trained on manually extracted acoustic features like mel-frequency cepstral coefficients. Recent reviews in this field have detailed the strengths and limitations of these conventional approaches. While deep learning techniques have begun to penetrate this area over the past few years, existing efforts often rely on highly preprocessed acoustic features, limiting their ability to capture the raw, intrinsic characteristics of music signals.

Residual Networks[7], a variant of CNNs, have addressed the vanishing gradient problem in deep network training, enabling the construction of deeper and more powerful models that excel at feature extraction from complex data. The GTZAN dataset[6], a widely used benchmark in MIR, provides a standardized collection of music samples across ten distinct genres, laying a solid foundation for evaluating classification models. Given the gaps in current music classification research—particularly the underutilization of advanced CNN architectures like ResNet and the over-reliance on manual feature engineering—an important research question emerges: Can ResNet-based CNN models effectively classify music genres using the GTZAN dataset, while reducing dependence on extensive preprocessing? This paper aims to address this question by developing and evaluating a ResNet-based approach for music genre classification on the GTZAN dataset.

II. CONVOLUTIONAL NEURAL NETWORK

A.Model Selection

Residual Network (ResNet) is selected as the core convolutional neural network architecture for music classification tasks. Different from traditional convolutional neural networks, ResNet addresses the gradient vanishing and performance degradation issues encountered in deep network training by introducing residual connections. This unique structural design allows the network to maintain stable training even with increased depth, enabling more effective extraction of complex hierarchical features from music audio data. For music genre classification based on the GTZAN dataset, ResNet’s strong feature learning capability aligns well with the demand for capturing subtle differences between diverse music genres, making it a suitable choice for the model framework of this study.

B.Network Architecture

The ResNet-based music classification network maintains the classic residual block structure while making appropriate adjustments to adapt to the characteristics of music data. The overall architecture consists of an input layer, a sequence of residual blocks, pooling layers, fully connected layers, and an output layer(depicted in Figure 1).

The input layer accepts log-mel spectrogram features of music audio . Each residual block contains two or three convolutional layers with the same number of filters, using 3×3 convolution kernels to capture local feature patterns of the spectrogram. The shortcut connection in the residual block skips one or more convolutional layers and directly adds the input of the block to the output after convolution and activation, which effectively alleviates the gradient vanishing problem by enabling the gradient to propagate smoothly through the shortcut path during backpropagation.

After the sequence of residual blocks, global average pooling is employed to reduce the dimensionality of the feature maps, retaining key feature information while reducing the number of model parameters. The fully connected layers following the pooling layer further integrate the extracted features, with dropout layers inserted between the fully connected layers to prevent overfitting. Finally, a softmax output layer with

10 neurons is used to output the classification probability distribution.

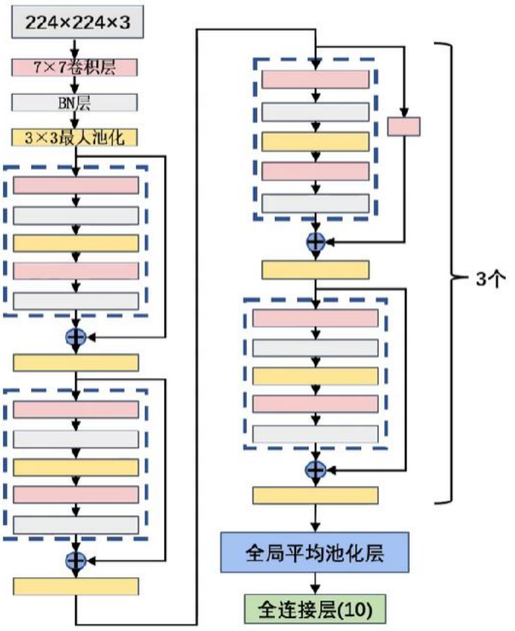

Figure 1 Visualization of a typical residual network architecture

C.Activation Function and Regularization

The rectified linear unit (ReLU) is adopted as the activation function for the convolutional layers and fully connected layers in the network. Its expression is f(x)=max(0, x), which offers advantages such as fast computation speed and efficient gradient propagation, avoiding the saturation problem of traditional sigmoid and tanh functions. This helps accelerate the convergence of the ResNet model during the training process with music data.To enhance the generalization ability of the model and mitigate overfitting, multiple regularization strategies are implemented. In addition to the dropout layers inserted in the fully connected layers, L2 weight decay is applied in the optimizer to constrain the magnitude of model parameters. During the data preprocessing stage, data augmentation operations such as random horizontal flipping, slight rotation, and color jitter are performed on the log-mel spectrogram data, which expands the effective size of the training dataset and reduces the risk of the model overfitting to specific training samples.D. Training StrategyThe training process of the ResNet model follows a systematic strategy tailored to the GTZAN dataset. The cross-entropy loss function is used to measure the difference between the predicted genre distribution and the true label distribution. The AdamW optimizer is selected for parameter updates, which combines the advantages of adaptive learning rate adjustment and weight decay, enabling more stable and efficient training.A dynamic learning rate adjustment strategy is employed during training: an initial learning rate of 0.001 is set, and the ReduceLROnPlateau scheduler is used to reduce the learning rate by half if the validation loss does not decrease for 3 consecutive epochs. This helps the model converge to a better local optimum. The batch size is set to 32, and the model is trained for a total of 20 epochs. To ensure the reproducibility of experimental results, a fixed random seed is used for initialization of model parameters, data shuffling, and other random operations.

III. RELATED WORK

Music genre classification, as a core task in music information retrieval, has attracted continuous attention from researchers. Early studies predominantly relied on traditional machine learning algorithms combined with manually engineered acoustic features. Among these features, mel-frequency cepstral coefficients were widely adopted for their ability to simulate human auditory perception, and were often used in conjunction with classifiers such as Gaussian mixture models and support vector machines to build classification systems. However, these methods heavily depend on professional domain knowledge for feature design, and their ability to capture complex hierarchical features in music is limited, leading to bottlenecks in classification performance when facing diverse music genres.

With the advancement of deep learning, convolutional neural networks have been gradually applied to music classification tasks, leveraging their powerful automatic feature extraction capabilities to reduce reliance on manual feature engineering. Early convolutional neural network models for music classification mainly used shallow network structures, extracting features from preprocessed audio representations such as mel spectrograms and then performing classification through fully connected layers(depicted in Figure 2).These attempts have demonstrated that CNN-based methods outperform traditional approaches in capturing local temporal-frequency features of music. However, shallow networks still struggle to model the deep semantic information of music, and their generalization ability is constrained when dealing with the rich stylistic variations in the GTZAN dataset.

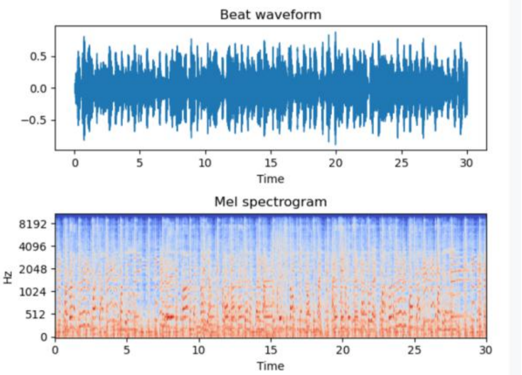

Figure 2 Beat wavform and extracted melspectrogram of raw audio

Residual networks, with their unique residual connection design, have overcome the challenges of gradient vanishing and performance degradation in deep network training, and have been successfully applied in various pattern recognition tasks. In the field of audio and music processing, relevant studies have shown that ResNet can effectively extract multi-scale and hierarchical features from audio spectrograms. For example, some works have used ResNet to classify environmental sounds and speech signals, achieving significant performance improvements. However, there is still room for exploration in the application of ResNet to GTZAN-based music genre classification—especially in optimizing the network structure to adapt to the characteristics of music spectrograms and improving the utilization of genre-related feature information.

In terms of data processing, log-mel spectrogram has become a mainstream input representation for music classification due to its ability to retain key temporal-frequency information of audio while conforming to the input format of convolutional neural networks. Tools such as LibROSA provide convenient support for extracting log-mel spectrogram features from raw audio, and related studies have confirmed that this feature representation is superior to traditional MFCCs in deep learning models. Existing works have laid a foundation for ResNet-based music classification, but further optimization is needed to better adapt to the specific characteristics of the GTZAN dataset and improve the accuracy and robustness of genre classification.

IV. MUSIC CLASSIFICATION

A. Datasets

The quality and scale of datasets are critical factors affecting the training effectiveness and generalization ability of deep learning models in music classification tasks. For music genre classification research, selecting a standardized and widely recognized dataset is essential to ensure the credibility and comparability of experimental results. This study adopts the GTZAN Music Genre Dataset as the experimental dataset, a classic benchmark in the field of music information retrieval. The dataset includes 10 music genres: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, and rock. Each genre consists of 100 audio clips, each lasting 30 seconds, with a sampling rate of 22050 Hz and a 16-bit depth. These audio clips cover the typical stylistic characteristics of each genre, laying a solid foundation for model training and performance evaluation.

During dataset preprocessing, the integrity and validity of the data are first verified. Referring to the evaluation criteria of the GTZAN dataset, potential issues such as duplicate clips, mislabeling, and audio distortion are checked. For a small number of clips with ambiguous labels, manual listening and confirmation are conducted to ensure consistency between the audio content and genre labels. Subsequently, the dataset is divided into a training set and a test set at an approximate ratio of 9:1. The training set contains 900 samples for model parameter learning and optimization, while the test set includes 100 samples to evaluate the generalization ability of the trained model, avoiding overfitting caused by the model's over-adaptation to training data.

B. Experiment Setup

The experimental setup is designed based on the characteristics of the ResNet model and the GTZAN dataset to ensure the effectiveness and rationality of the training process. In terms of data preprocessing, the LibROSA tool is used to extract log-mel spectrogram features from original audio clips. The specific parameters are set as follows: the audio is resampled to 22050 Hz, a Hamming window with a size of 512 is adopted, the hop length is set to 256, and 128 mel-bands are selected to generate spectrograms. The generated log-mel spectrograms are normalized to the range [0, 1] and then converted into a 3-channel image-like format to match the input requirements of the ResNet model. Additionally, data augmentation operations such as random horizontal flipping, random rotation within ±10 degrees, and brightness adjustment are applied to the training set to expand the diversity of training data and enhance the model's robustness.

In terms of model configuration, ResNet18 is selected as the base model and appropriately adjusted for music classification tasks. The final fully connected layer of the original ResNet18 is replaced with a custom output layer: after the global average pooling layer, a dropout layer with a dropout rate of 0.5 is added, followed by two fully connected layers and a ReLU activation function. Finally, a softmax layer with 10 neurons is used to output the classification probability of each genre. During training, the cross-entropy loss function is employed to measure the difference between predicted values and true labels, and the AdamW optimizer is selected with an initial learning rate of 0.001 and a weight decay of 1e-4 to prevent overfitting. The ReduceLROnPlateau learning rate scheduler is used to dynamically adjust the learning rate: if the validation loss does not decrease for 3 consecutive epochs, the learning rate is reduced to 50% of its current value. The batch size is set to 32, and the model is trained for a total of 20 epochs. All experiments are conducted on a GPU device to accelerate the training process.

C. Result

The performance of the ResNet18-based music classification model is evaluated from multiple perspectives, including training process curves, test set accuracy, genre-specific classification effects, and precision-recall metrics. Figure 3 and Figure 4 show that the model's training loss and validation loss continuously decrease in the early training stage, with the training loss dropping from 1.9179 to 0.7016 and the training accuracy increasing from 30.37% to 77.86% over 20 epochs. The validation accuracy fluctuates during the training process but reaches a maximum of 71.00%, indicating that the model has good convergence performance. The data augmentation and regularization strategies effectively mitigate overfitting, though slight fluctuations in validation performance reflect the model's sensitivity to certain genre-specific feature variations.

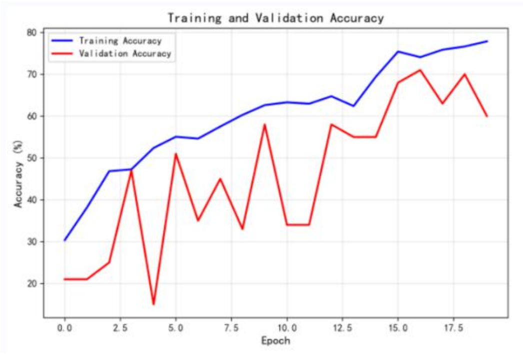

Figure 3: Training and Test Accuracy Plot

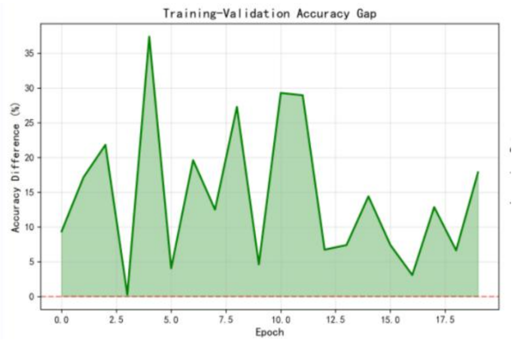

Figure 4: Training-Validation Accuracy Difference Plot

On the test set, the model achieves an overall classification accuracy of 60.00%. The genre-specific classification results reveal significant differences in performance across different genres. The classical genre achieves perfect scores in precision, recall, and F1-score, benefiting from its distinct spectral characteristics—gentle spectral distribution and slow tempo—that are easily distinguishable from other genres. Genres such as country and metal also perform well, with F1-scores of 0.71 and 0.70 respectively. In contrast, genres like jazz and rock have relatively low F1-scores, mainly due to their overlapping temporal and spectral features with other genres. For example, jazz shares similar rhythmic patterns with blues, while rock has stylistic overlaps with reggae, leading to partial misclassifications.

The classification report(depicted in Figure 2 and Table 1) shows that the model's macro-average precision is 0.71, macro-average recall is 0.60, and macro-average F1-score is 0.58. The weighted average precision and F1-score are consistent with the macro-average, indicating that the model's performance is balanced across different genres without significant bias toward majority classes. The confusion matrix further confirms that the main misclassifications occur between stylistically similar genres: blues is occasionally misclassified as jazz, reggae as rock, and hiphop as disco. These misclassifications align with the inherent similarities in the musical structures and spectral features of these genres.

Table 1 Classification Performance Metrics of GTZAN Music Genre Excerpts

	precision	recall	f1-score
blues	0.54	0.70	0.61
classical	1.00	1.00	1.00
country	0.86	0.60	0.71
disco	0.50	0.80	0.62
hiphop	0.37	0.70	0.48
jazz	1.00	0.20	0.33
metal	0.62	0.80	0.70
pop	0.46	0.60	0.52
reggae	0.80	0.40	0.53
rock	1.00	0.20	0.33
accuracy	---	---	0.60
macro avg	0.71	0.60	0.58
weighted avg	0.71	0.60	0.58

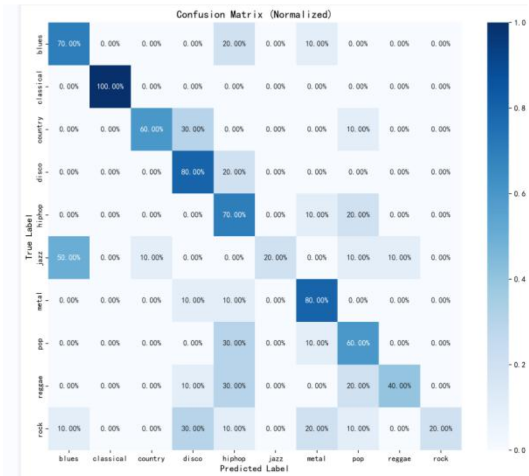

Figure 5 Normalized Confusion Matrix

Overall, the experimental results demonstrate that the ResNet18 model can effectively extract discriminative features from the log-mel spectrograms of music audio, achieving reasonable classification performance on the GTZAN dataset. The model's strong performance on distinct genres and relatively weak performance on similar genres provides valuable insights for future improvements. These include optimizing the network structure to enhance the capture of fine-grained genre features, expanding the dataset to include more diverse samples, and adjusting feature extraction parameters to better distinguish between stylistically similar genres.

V. SUMMARY

This study focuses on exploring the application of Residual Networks (ResNet) in music genre classification tasks, with the GTZAN dataset as the experimental benchmark. The core objective is to verify whether ResNet can effectively reduce reliance on manual feature engineering while achieving reliable classification results for music genres. Through systematic experimental design, log-mel spectrogram features were extracted from raw audio data using the LibROSA tool, and a ResNet18-based classification model was constructed with optimized network structure, activation functions, and training strategies. After 20 epochs of training, the model achieved an overall test accuracy of 60.00% on the GTZAN dataset, with exceptional performance on distinct genres such as classical music (F1-score of 1.00), demonstrating ResNet's strong capability in extracting discriminative features from audio spectrograms.

However, the experimental results also reveal limitations of the proposed approach. The classification performance varies significantly across genres, with relatively low F1-scores for stylistically similar genres like jazz and rock, mainly due to overlapping temporal and spectral features that are difficult to distinguish. Additionally, although data augmentation and regularization strategies were adopted, slight fluctuations in validation accuracy during training indicate potential room for improvement in the model's generalization ability. These findings suggest that while ResNet provides a viable solution for music genre classification, further optimizations are needed to address the challenges posed by genre similarity and limited dataset diversity.

Looking forward, future research can be expanded in several directions. First, optimizing the network architecture—such as adopting deeper ResNet variants (e.g., ResNet50) or integrating attention mechanisms—to enhance the capture of fine-grained genre features. Second, enriching the dataset with more diverse music samples and addressing label ambiguity to reduce the impact of data bias on classification results. Third, exploring advanced feature extraction methods or fusing multi-modal features (e.g., combining spectral features with rhythmic features) to improve the distinguishability between similar genres. Overall, this study confirms the feasibility of ResNet in music genre classification and provides a solid foundation for subsequent research on more robust and accurate music classification systems.

REFERENCES

[1] Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural net- works," in Advances in Neural Information Processing Systems, 2012, pp. 1097–1105.

[2] Piczak,Karol J. . "Environmental sound classification with convolutional neural networks."2015 IEEE 25th International *Workshop on Machine Learning for Signal Processing* (MLSP)IEEE,2015.

[3] Wang, Xiyue , et al. "Automatic Hypernasality Detection in Cleft Palate Speech Using CNN." *Circuits Systems and Signal Processing* 15(2019).

[4] S. Dieleman, P. Brakel, and B. Schrauwen, "Audio-based music classification with a pretrained convolutional network," in Proceedings of the 12th International Society for Music Information Retrieval (ISMIR) conference, 2011, pp. 669–674.

[5] Van den Oord, S. Dieleman, and B. Schrauwen, “Deep contentbased music recommendation,” in Advances in Neural Information Processing Systems, 2013, pp. 2643–2651.

[6] Sturm, Bob L. . "An analysis of the GTZAN music genre dataset." *ACM Workshop on Music Information Retrieval with User Centered and Multimodal Strategies* ACM, 2012.

[7] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), Jun. 2016, pp. 770–778.